BME 6938 · ADVANCED BIOMEDICAL AI · SPRING 2027

LECTURE 5-1

Post-Training and Reasoning

From pretrained model to clinical assistant



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Where We Are

Last week: what is inside the model. Today: how training turns it into an assistant.

Week 4	Week 5: today	Week 6
Inside modern LLMs	**Post-training and reasoning**	**Self-supervised learning and imaging**
The block, the KV cache, mixture of experts	Fine-tuning, LoRA, preference tuning, reasoning models	Foundation models for biomedical data

A Pretrained Model Only Continues Text

Given the start of a discharge summary, it writes more note, about a different patient.

Discharge summary: The patient denies chest pain.



Qwen3-0.6B-Base continues (real output)

The patient is a 65-year-old woman with a history of hypertension and diabetes.

She has been experiencing fatigue and shortness of breath for the …

Pretraining teaches one thing: continue the text.

Ask a Question, Get a Quiz

The base model treats a question as text to continue, and keeps inventing questions.

Note: The patient denies chest pain.

Question: Does the patient report chest pain?

↓

Qwen3-0.6B-Base continues (real output)

(Yes/No)

Answer: No

Question: Does the patient report shortness of breath? (Yes/No)

Answer: Yes

Question: …

Post-Trained, but Wrong

The post-trained model answers like an assistant. Here, its answer is wrong.

Note: The patient denies chest pain.

Question: Does the patient report chest pain?

↓

Post-trained Qwen3-0.6B (real output)

“Yes, the patient reports chest pain.”

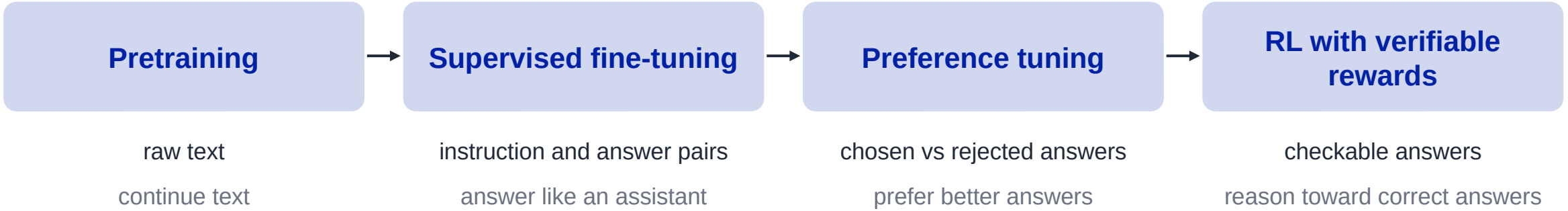
wrong: the note says “denies”

Today: three ways to fix this negation error,

and what each one costs.

Four Stages After the Architecture

Same model and same network; each stage changes what the model is trained on.



Sections 1 to 4 of today follow these stages:
fine-tuning · fine-tuning cheaply (LoRA) · preference tuning · reasoning



Supervised Fine-Tuning

Instruction and Answer Pairs in a Chat Template

Special markers show where the user's turn and the assistant's turn begin and end.

<|im_start|>user	role marker
Note: The patient denies chest pain. Question: Does the patient report chest pain?	instruction
<|im_end|> <|im_start|>assistant	role marker
No. The note says the patient denies chest pain.	answer
<|im_end|>	end marker

Same Loss, Counted Only on the Answer

The next-token loss is computed on the answer tokens; the instruction is context.

Note	:	The	patient	denies	chest	pain	.	Question	:	Does	…	?

instruction: not counted in the loss

No	.	The	note	says	the	patient	denies	chest	pain	.

answer: counted in the loss

loss = average over answer tokens of $-\log p(\text{next token} \mid \text{everything before it})$

Animation: From Pretrained Model to Clinical Assistant

Animation clip: week05-clip1-post-training.mp4

Silent version embedded (click to play). The narrated version is in the course repository.

Same Architecture, Same Loss, Different Data

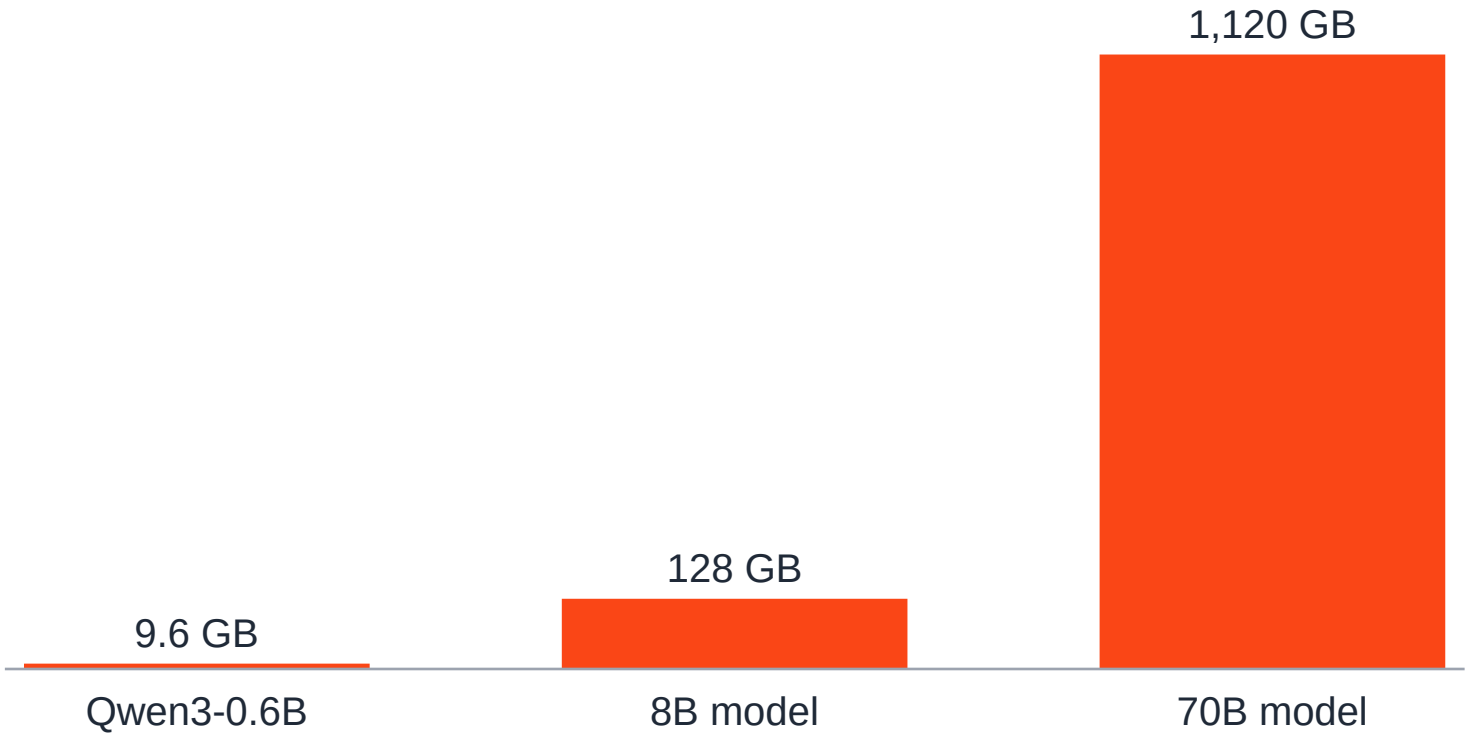
architecture	decoder blocks	→	unchanged
loss	next-token cross-entropy	→	unchanged
data	raw text	→	**instruction and answer pairs**

2 LoRA

What Full Fine-Tuning Costs

With the Adam optimizer, every parameter needs about 16 bytes of memory.

weight + gradient + 2 Adam states = 4 numbers × 4 bytes = 16 bytes per parameter



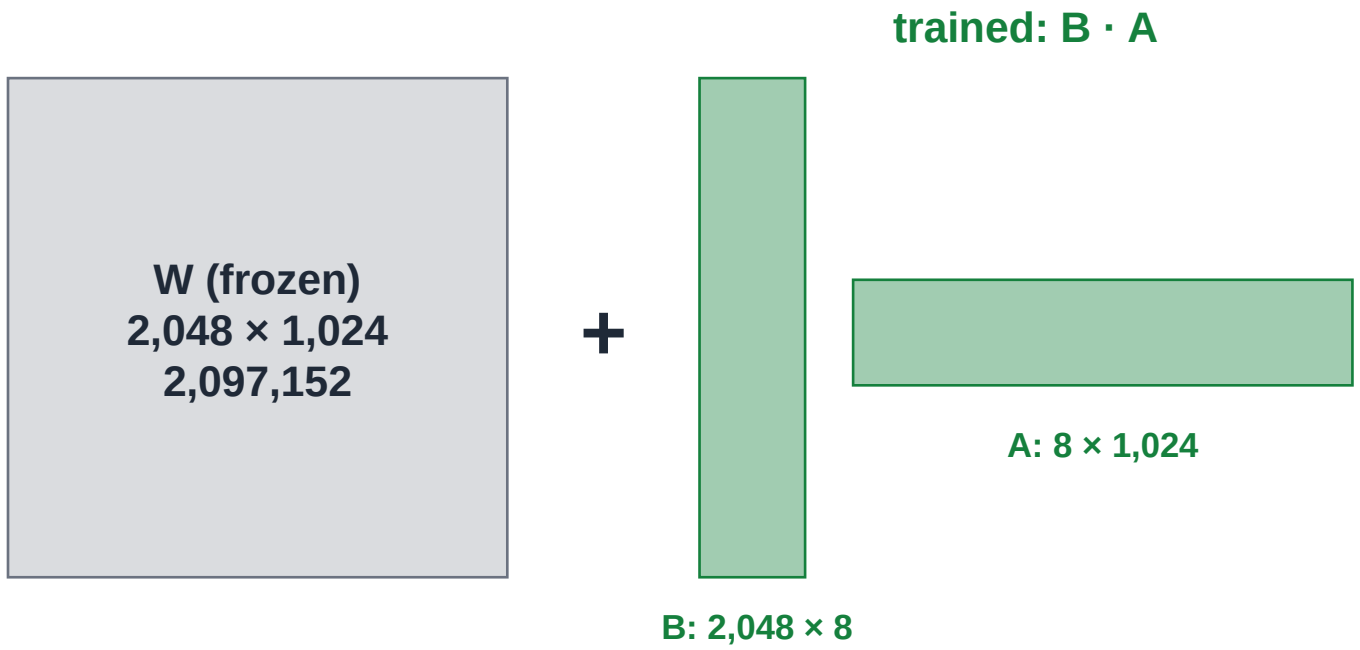
memory for full fine-tuning (weights, gradients, Adam states)

one GPU

80 GB is a large data-center GPU

LoRA: Freeze W, Train Two Thin Matrices

The output becomes W·x + B·A·x. Only B and A are trained.



output = W·x + B·A·x

A: 1,024 → 8 numbers
B: 8 → 2,048 numbers

The pretrained knowledge in W is untouched; B·A learns the task-specific change.

Counting the Weights

Rank-8 LoRA trains about 1.2% of one matrix and about 0.4% of the model.

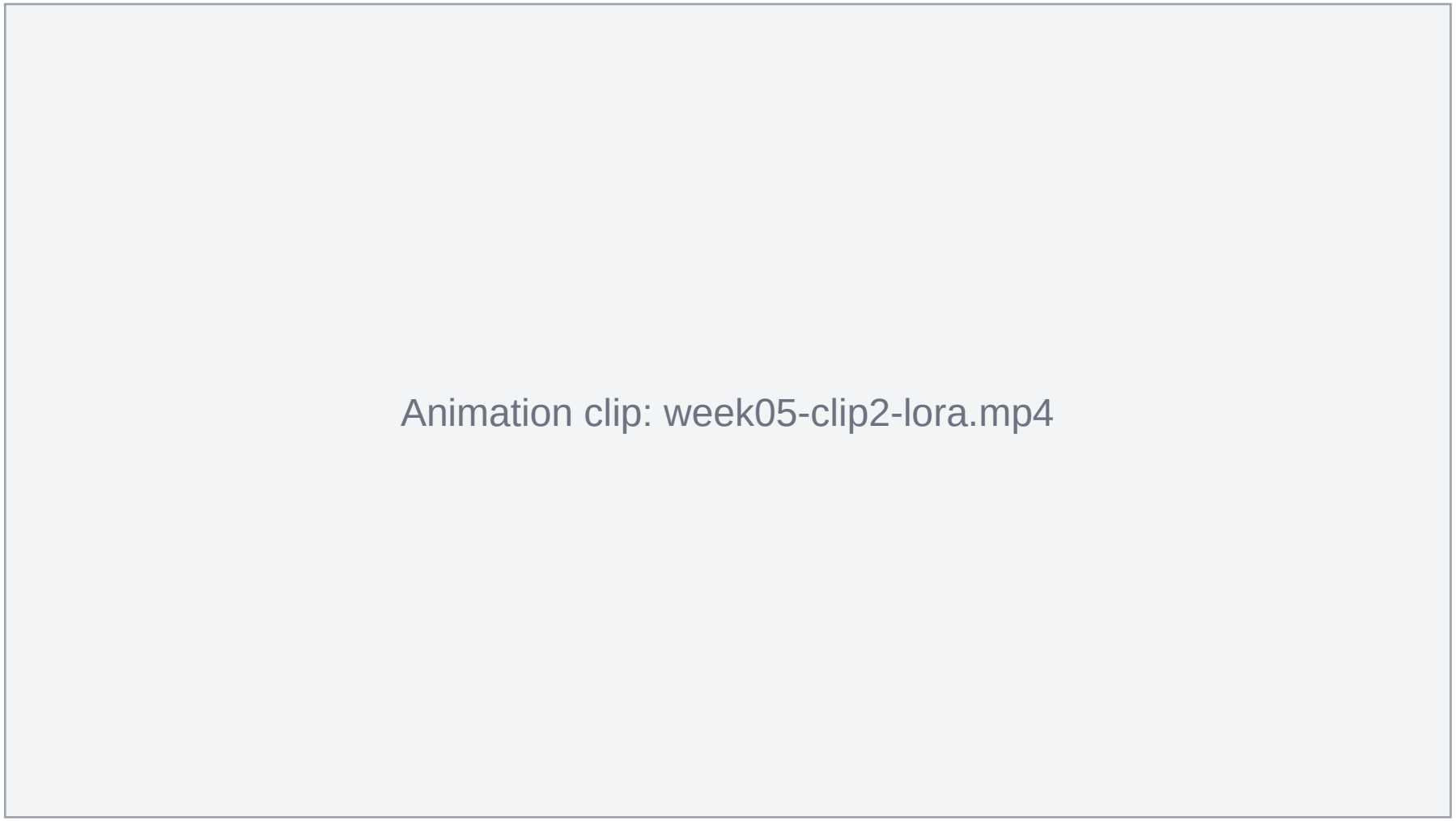
W, one query projection	**2,097,152**
B + A for that matrix	**24,576 (1.2%)**
all q, k, v, o in 28 layers	**2,293,760 (≈ 0.4% of the model)**
adapter file (16-bit)	**about 4.6 MB**

B Starts at Zero

At the first step, LoRA changes nothing: training starts exactly from the original model.

initialization	A random, B = 0
first step	$B \cdot A \cdot x = 0$, so output = $W \cdot x$
training	only B and A change; W stays frozen

Animation: LoRA

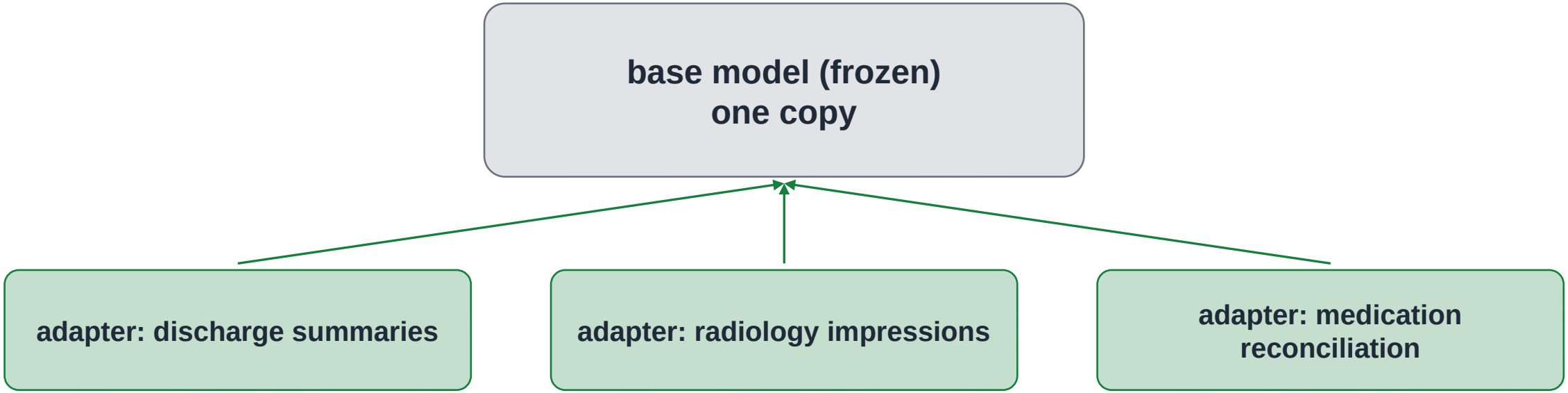


Silent version embedded (click to play). The narrated version is in the course repository.

One Base Model, Many Clinical Adapters

Swap a 4.6 MB adapter instead of copying a whole model for every task.

Notebook: a 12-second LoRA run on CPU fixes the chest-pain negation error.





Preference Tuning

Preference Data: Chosen and Rejected

A clinician marks one answer as better. The model currently prefers the wrong one.

Note: The patient denies chest pain.

Question: Does the patient report chest pain?

chosen

“No. The note says the patient denies chest pain.”

log-probability now: −21.37

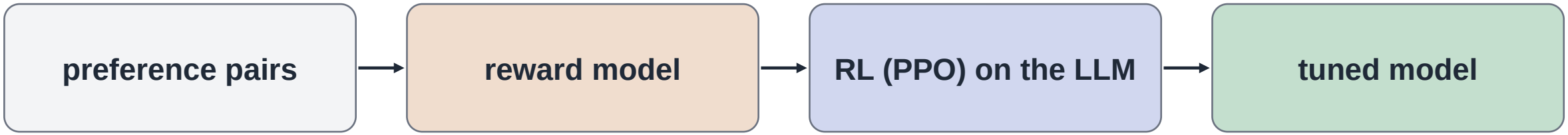
rejected

“Yes, the patient reports chest pain.”

log-probability now: −1.12

RLHF: A Reward Model, Then Reinforcement Learning

The classic method: learn a reward from preferences, then optimize it.



learns a score:
chosen above rejected

answers that score high
become more likely

cost: a separate reward model, and an RL step that can be unstable

DPO: Skip the Reward Model

Compare the model in training with a frozen copy of itself.

$$\text{loss} = -\log \sigma\left(\beta \cdot [\ \text{chosen gain} - \text{rejected gain}\ ] \right)$$

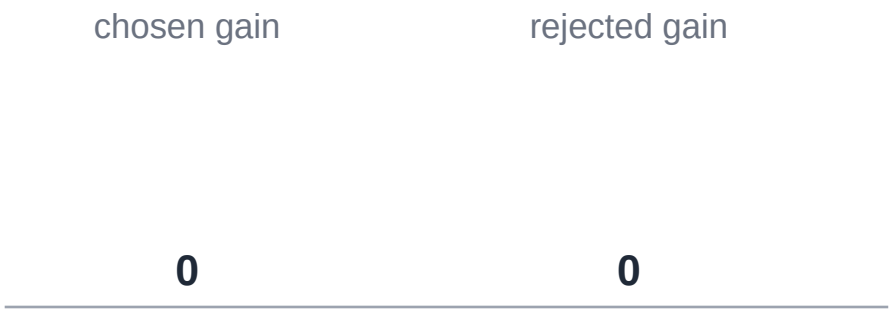
gain	log p(answer | model in training) − log p(answer | frozen copy)
β	a small number (here 0.1): how far the model may move from the frozen copy
σ	the sigmoid, which maps the scaled gap to a number between 0 and 1

DPO in Numbers

As the chosen gain pulls ahead of the rejected gain, the loss falls.

at the start

loss = 0.69



after training

loss = 0.31



Animation: Preference Tuning: RLHF and DPO

Animation clip: week05-clip3-preference-tuning.mp4

Silent version embedded (click to play). The narrated version is in the course repository.

What Preference Data Teaches

The model learns whatever the preferences reward, including bad habits.

In the notebook

a 21-second DPO run fixes the error
it carries over to headache and
leg swelling (not in training)
“reports” notes still get “Yes”

The risk

if the cautious answer always wins,
the model learns to hedge:
“further assessment is recommended”

Reward answers that are correct and useful, not answers that merely sound careful.



Reasoning Models

A Reasoning Model Thinks Before It Answers

It writes its reasoning inside <think> tags, then gives the answer.

A 58-year-old man has crushing chest pain.
ECG: ST elevation in leads II, III, and aVF.
What is the most likely occluded artery?

A) left anterior descending
B) right coronary
C) left circumflex
D) left main

<think> (illustrative trace)

ST elevation in II, III, and aVF
points to the inferior wall.
The inferior wall is usually supplied
by the right coronary artery.

</think> Answer: B

The reasoning is part of the output: it costs tokens, and it can be read and checked.

Rewards a Program Can Check

DeepSeek-R1 learned to reason from rule-based rewards, without a reward model.

Pretraining	Supervised fine-tuning	Preference tuning	**RL with verifiable rewards**
raw text	instruction and answer pairs	chosen vs rejected answers	checkable answers
continue text	answer like an assistant	prefer better answers	reason toward correct answers

Two rewards

accuracy: 1 if the final letter is correct

format: 1 if the reasoning is inside <think> tags

R1-Zero and R1

R1-Zero: RL only, from the base model

R1: a small cold-start SFT step, then RL

GRPO: Compare Each Answer with Its Own Group

Sample a group of answers to one question; score each against the group.

sample 1 Answer: B	sample 2 Answer: A	sample 3 Answer: C	sample 4 Answer: D
reward 1	**reward 0**	**reward 0**	**reward 0**
+1.73	**−0.58**	**−0.58**	**−0.58**

advantage = (reward − group mean) ÷ group std mean 0.25, std 0.433

4 here;
R1 used 16
per question

answers above their group average become more likely; the others less likely

Animation: Reasoning Models

Animation clip: week05-clip4-reasoning.mp4

Silent version embedded (click to play). The narrated version is in the course repository.

Thinking at Test Time

Thinking longer can fix an error, at the cost of more tokens and more time.

Note: The patient denies chest pain.

Question: Does the patient report chest pain?

Qwen3-0.6B, thinking off

“Yes, the patient reports chest pain.”

9 tokens · 0.9 s · wrong

Qwen3-0.6B, thinking on

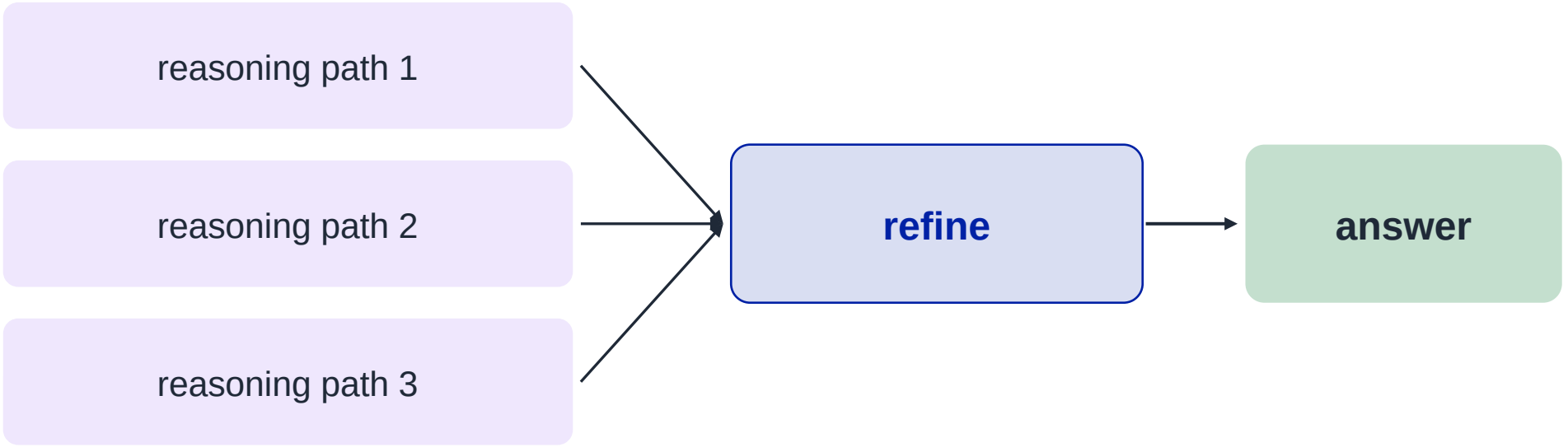
“The patient does not report chest pain.”

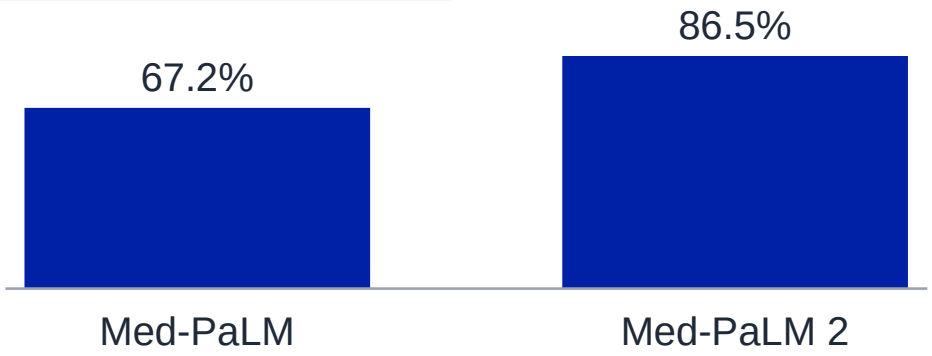
119 tokens · 11.4 s · correct

real outputs, CPU, greedy decoding

Med-PaLM 2: Ensemble Refinement

Sample several reasoning paths, then condition on all of them to write one answer.





MedQA accuracy
(stronger base model, medical
fine-tuning, and refinement together)

A Fluent Rationale Is Not Proof

Thinking can also lead a model away from the right answer.

thinking off

answer: B, right coronary

correct

thinking on

“ST elevation in the inferior leads is most commonly due to a blockage in the left anterior descending artery.”

after ~1,500 tokens: still unsure, leaning to A

Check the answer, not the eloquence.
Week 12: evaluating reasoning and its explanations.

Recap: From Pretrained Model to Clinical Assistant

Fine-tuning	Instruction and answer pairs, same loss, counted only on the answer.
LoRA	Freeze W, train B·A: about 0.4% of the weights, a 4.6 MB adapter per task.
Preferences	RLHF learns a reward; DPO compares with a frozen copy. Data decides what is learned.
Reasoning	Rewards a program can check, GRPO, thinking at test time. Fluent is not proof.

Before Thursday

Read	Guo et al., DeepSeek-R1 (Nature 2025) and Singhal et al., Med-PaLM 2 (Nature Medicine 2025).
Write	Reading responses on Canvas before 1:00 PM Thursday.
Explore	Week 5 reference notebook (ungraded): LoRA, the DPO loss, and GRPO from scratch; short runs that fix the negation error.
Next week	Self-supervised learning and imaging foundation models.